Coordination and response

60 questions: 30 multiple choice, 30 structured.

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1 · Part 1 Core Coordination and response [1]

Which structures make up the central nervous system (CNS) of a mammal?

A the brain and all the nerves of the body

B the brain and the sense organs

C the brain and the spinal cord

D the spinal cord and the nerves that leave it

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2 · Part 1 Core Coordination and response [1]

In what form does information travel along a neurone?

A electrical impulses

B hormones

C heat energy

D oxygen molecules

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3 · Part 1 Core Coordination and response [1]

Someone accidentally touches a hot pan and quickly pulls their hand away.

Which sequence shows the order in which the structures in this reflex arc are used?

A receptor → motor neurone → relay neurone → sensory neurone → effector

B effector → sensory neurone → relay neurone → motor neurone → receptor

C receptor → relay neurone → sensory neurone → motor neurone → effector

D receptor → sensory neurone → relay neurone → motor neurone → effector

4 · Part 1 Core Coordination and response [1]

Which statement describes a reflex action?

A a learned response that becomes faster with practice

B an automatic and rapid response of a muscle or gland to a stimulus

C a slow response that is controlled by conscious decisions made in the brain

D a response that does not involve any neurones

5 · Part 1 Core Coordination and response [1]

When a speck of dust touches the surface of the eye, the person blinks. This is a reflex action.

Which structure is the effector in this reflex?

A touch receptors on the surface of the eye

B muscles that close the eyelid

C the sensory neurone

D the relay neurone

6 · Part 1 Core Coordination and response [1]

What is a synapse?

A a receptor cell in a sense organ

B the long fibre of a neurone that carries impulses

C a bundle of many neurones outside the brain and spinal cord

D a junction between two neurones

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7 · Part 1 Supplement Coordination and response [1]

The statements describe four events at a synapse.

1 Neurotransmitter molecules diffuse across the synaptic gap.

2 An impulse arrives and stimulates the release of neurotransmitter molecules from vesicles.

3 An impulse is stimulated in the next neurone.

4 Neurotransmitter molecules bind with receptor proteins on the next neurone.

What is the correct order of these events?

A 2 → 1 → 4 → 3

B 2 → 4 → 1 → 3

C 1 → 2 → 4 → 3

D 4 → 2 → 1 → 3

8 · Part 1 Supplement Coordination and response [1]

Impulses can pass across a synapse in one direction only.

Which statement explains this?

A Receptor proteins are found on the membranes of both neurones at the synapse.

B Both neurones at the synapse release neurotransmitter into the synaptic gap.

C Only the neurone before the gap has vesicles of neurotransmitter, and only the neurone after the gap has receptor proteins.

D Neurotransmitter molecules are too large to diffuse back across the synaptic gap.

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9 · Part 1 Core Sense organs [1]

Which row correctly matches a sense organ with a stimulus that its receptor cells respond to?

	sense organ	stimulus
A	ear	light
B	eye	chemicals
C	skin	sound
D	tongue	chemicals

10 · Part 1 Core Sense organs [1]

Which row correctly matches a part of the eye with its function?

	part of the eye	function
A	cornea	refracts light
B	iris	focuses light on to the retina
C	optic nerve	contains light receptors
D	lens	controls how much light enters the pupil

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11 · Part 1 Core Sense organs [1]

Someone walks out of a dark room into bright sunlight.

Which row shows the change in the diameter of their pupils and the effect of this change on the amount of light entering each eye?

	change in pupil diameter	**effect of this change on the amount of light entering the eye**
A	increases	more light enters
B	increases	less light enters
C	decreases	less light enters
D	decreases	more light enters

12 · Part 1 Supplement Sense organs [1]

The light in a room is switched off, leaving only a dim lamp.

Which row shows what happens to the muscles of the iris?

	circular muscles	**radial muscles**
A	contract	contract
B	relax	contract
C	contract	relax
D	relax	relax

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13 · Part 1 Supplement Sense organs [1]

Which row describes the eye when it is focused on a near object?

	ciliary muscles	suspensory ligaments	shape of lens
A	contracted	loose (low tension)	more rounded
B	relaxed	tight (high tension)	thinner
C	contracted	tight (high tension)	thinner
D	relaxed	loose (low tension)	more rounded

14 · Part 1 Supplement Sense organs [1]

In very dim light, a person can see the shapes of objects but cannot tell what colour they are.

Which statement explains this?

A Cones are more sensitive to light than rods, so cones work best in dim light.

B There are three types of rod, each absorbing light of a different colour.

C Rods are found only in the centre of the retina.

D Rods are more sensitive to light than cones, but rods do not give colour vision.

15 · Part 1 Supplement Sense organs [1]

Which statement describes the distribution of rods and cones in the human retina?

A Cones are spread evenly over the whole retina.

B Rods are most concentrated at the centre of the retina.

C Cones are most concentrated at the centre of the retina, and rods are more common away from the centre.

D Rods and cones are both present in large numbers where the optic nerve leaves the eye.

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16 · Part 1 Core Coordination and response [4]

The mammalian nervous system is made up of the central nervous system (CNS) and the peripheral nervous system (PNS).

(a) State the two parts of the CNS. [2]

(b) State what makes up the PNS. [1]

(c) Describe the role of the nervous system in the body. [1]

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17 · Part 1 Core Coordination and response [5]

While walking barefoot, Maya steps on a sharp pin. Her foot is lifted off the pin before she is aware of any pain. This is a reflex action.

(a) State where the receptor for this reflex is found. [1]

(b) Identify the type of neurone that carries impulses from the receptor to the spinal cord. [1]

(c) Identify the type of neurone, found in the spinal cord, that passes impulses on to the motor neurone. [1]

(d) State the effector in this reflex and describe its response. [2]

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18 · Part 1 Core Coordination and response [4]

Reflex actions automatically and rapidly coordinate stimuli with the responses of effectors. Effectors can be muscles or glands.

(a) Explain why it is an advantage that the reflex that pulls a hand away from a flame is automatic and rapid. [2]

(b) When food is placed in the mouth, saliva is released straight away, without the person thinking about it.

(i) Identify the effector in this reflex. [1]

(ii) State the response of this effector. [1]

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19 · Part 1 Core Coordination and response [6]

Tapping the tendon just below the kneecap makes the lower leg kick forward. This is the knee-jerk reflex. The mean time between the tap and the kick for one student was 0.05 s.

The same student then took part in a ruler-catching test. A partner held a ruler and let go of it without warning, and the student caught it as quickly as possible. The table shows the time taken to catch the ruler in five trials.

trial	time to catch ruler / s
1	0.21
2	0.19
3	0.24
4	0.20
5	0.21

(a) Calculate the mean time to catch the ruler. [2]

(b) Calculate how many times longer the mean ruler-catching time is than the mean knee-jerk time. [1]

(c) Explain why the knee-jerk response is faster than catching the ruler. [2]

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(d) State the type of effector that causes the lower leg to kick forward. [1]

20 · Part 1 Core Coordination and response [4]

When a finger touches a sharp object, the hand is pulled away. The pathway for this reflex is:

receptor in the skin → sensory neurone → relay neurone → motor neurone → muscle in the arm

(a) State the form in which information travels along each neurone. [1]

(b) State what is meant by a synapse. [1]

(c) Determine the number of synapses between neurones in this pathway. [1]

(d) State the part of the central nervous system where the relay neurone in this pathway is found. [1]

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21 · Part 1 Supplement Coordination and response [4]

At a synapse, the end of one neurone lies very close to the next neurone, but the two neurones do not touch.

(a) State the name of the space between the two neurones at a synapse. [1]

(b) State the name of the molecules that are stored in vesicles at a synapse. [1]

(c) State in which of the two neurones the vesicles are found. [1]

(d) State where the receptor proteins are found at a synapse. [1]

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22 · Part 1 Supplement Coordination and response [5]

In the spinal cord, a sensory neurone forms a synapse with a relay neurone.

(a) Describe the events at this synapse that lead to an impulse in the relay neurone. [4]

(b) Suggest how the time taken for an impulse to pass across a synapse would change if the synaptic gap were much wider. Give a reason for your answer. [1]

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23 · Part 1 Supplement Coordination and response [6]

Scientists are testing two drugs on synapses between neurones.

Drug P is a molecule with a shape very similar to the neurotransmitter. Drug P binds to the receptor proteins on the next neurone but does not stimulate an impulse. It stays bound for a long time.

Drug Q stops the vesicles from releasing neurotransmitter.

(a) Suggest why drug P is able to bind to the receptor proteins. [2]

(b) Suggest and explain the effect of drug P on the transmission of impulses across the synapse. [2]

(c) Explain why drug Q stops impulses being stimulated in the next neurone. [2]

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24 · Part 1 Supplement Coordination and response [5]

Three neurones, X, Y and Z, are joined in a chain by two synapses. In the body, impulses normally pass from X to Y and then from Y to Z.

In an experiment, a scientist stimulates neurone Y at a point halfway along its length. The impulse produced travels in both directions along neurone Y, reaching both of its synapses. The scientist records whether impulses are produced in neurone X and in neurone Z.

(a) Predict whether an impulse will be produced in neurone X and whether an impulse will be produced in neurone Z. [2]

(b) Explain your predictions. [3]

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25 · Part 1 Core Sense organs [4]

The eyes, ears, nose, tongue and skin are sense organs.

(a) Describe what is meant by a sense organ. [2]

(b) The skin contains several kinds of receptor cell. Give two different stimuli that receptor cells in the skin respond to. [2]

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26 · Part 1 Core Sense organs [5]

Light from a tree passes into the eye of an observer and the observer sees the tree.

(a) Identify the transparent part at the front of the eye that first refracts light entering the eye. [1]

(b) Identify the part of the eye that controls how much light enters the pupil. [1]

(c) Identify the part of the eye that focuses the light on to the retina. [1]

(d) After an injury, a patient's retina still works normally but their optic nerve is badly damaged. Explain why the patient cannot see with this eye. [2]

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27 · Part 1 Core Sense organs [5]

The diameter of a volunteer's pupil was measured at different light intensities. The results are shown in the table.

light intensity / lux	pupil diameter / mm
10	7.0
100	5.5
500	4.0
1000	3.0
2000	2.5

(a) Describe the relationship between light intensity and pupil diameter shown in the table. [2]

(b) Calculate the percentage decrease in pupil diameter when the light intensity increases from 10 lux to 1000 lux. Give your answer to the nearest whole number. [2]

(c) Explain why it is an advantage for the pupil diameter to be small in bright light. [1]

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28 · Part 1 Supplement Sense organs [5]

The iris contains two sets of muscles: circular muscles and radial muscles.

(a) Ravi is sitting in a brightly lit room when the main lights are switched off. Explain how the muscles of his iris cause a change in his pupils. [3]

(b) The circular and radial muscles are described as antagonistic. Explain what this means. [2]

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29 · Part 1 Supplement Sense organs [6]

Leila is reading a book held close to her face. She then looks up at a mountain far away. Her eyes change so that she sees the mountain clearly. This is called accommodation.

(a) State the part of the eye that changes shape during accommodation. [1]

(b) Explain the changes in Leila's eyes that allow her to see the distant mountain clearly after reading the book. [4]

(c) As people get older, the lens becomes stiffer and cannot easily become more rounded. Suggest why many older people find it difficult to read small print held close to their eyes. [1]

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30 · Part 1 Supplement Sense organs [6]

Scientists counted the rods and cones at five positions, P to T, along a line across a human retina. Position R is the centre of the retina.

position	distance from centre of retina / mm	rods / thousands per mm^2	cones / thousands per mm^2
P	10 (towards the ear)	90	5
Q	3 (towards the ear)	150	8
R	0	0	150
S	4 (towards the nose)	0	0
T	10 (towards the nose)	80	5

(a) Describe the distribution of cones shown in the table. [2]

(b) Suggest why there are no rods or cones at position S. [1]

(c) Astronomers know that a very dim star is easier to see if they look slightly to one side of it rather than straight at it. When looking straight at the star, its light falls on position R. Use the data to explain this observation. [2]

(d) Outline how cones allow humans to see in colour. [1]

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1 · Part 2 Core Hormones [1]

Which statement describes a hormone?

A an enzyme, produced by a gland, that travels along neurones to an effector

B a chemical substance, produced by a gland and carried by the blood, that alters the activity of one or more specific target organs

C an electrical impulse, produced by a receptor, that alters the activity of muscles and glands

D a chemical substance, produced by a gland and carried by the blood, that alters the activity of every organ in the body equally

2 · Part 2 Core Hormones [1]

Late at night, a man walking home hears a dog growling close behind him. His adrenal glands secrete adrenaline.

Which row shows the effects of adrenaline on the man?

	breathing rate	**heart rate**	**pupil diameter**
A	increases	decreases	increases
B	decreases	increases	decreases
C	increases	increases	decreases
D	increases	increases	increases

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3 · Part 2 Core Hormones [1]

Which row compares hormonal control with nervous control?

	speed of action of hormonal control	duration of effect of hormonal control
A	slower than nervous control	longer than nervous control
B	faster than nervous control	longer than nervous control
C	slower than nervous control	shorter than nervous control
D	faster than nervous control	shorter than nervous control

4 · Part 2 Supplement Hormones [1]

Which two hormones are both secreted by the pancreas?

A adrenaline and insulin

B adrenaline and glucagon

C glucagon and insulin

D oestrogen and insulin

5 · Part 2 Supplement Hormones [1]

Just before the start of a race, a sprinter secretes adrenaline.

Which statement explains how adrenaline helps the sprinter's muscle cells to release more energy during the race?

A It increases the blood glucose concentration and increases the heart rate, so more glucose and oxygen are delivered to the muscle cells for respiration.

B It decreases the blood glucose concentration by converting glucose to glycogen, so the muscle cells store more energy.

C It increases the heart rate, so more carbon dioxide is delivered to the muscle cells for respiration.

D It decreases the heart rate, so blood stays in the muscles for longer and more glucose diffuses into the muscle cells.

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6 · Part 2 Core Homeostasis [1]

What is homeostasis?

A the coordination of responses to stimuli in the external environment by the nervous system

B the maintenance of a constant internal environment

C the removal of the waste products of metabolism from the body

D the maintenance of a constant external environment around an organism

7 · Part 2 Core Homeostasis [1]

After a meal containing a lot of starch, the blood glucose concentration of a healthy person rises above normal.

Which row identifies the hormone that returns the blood glucose concentration to normal, and its effect?

	hormone	**effect on blood glucose concentration**
A	adrenaline	increases it
B	insulin	increases it
C	testosterone	decreases it
D	insulin	decreases it

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8 · Part 2 Supplement Homeostasis [1]

Which statement describes homeostatic control by negative feedback?

A When a condition moves away from its set point, the body responds in a way that moves the condition even further from the set point.

B When a condition moves away from its set point, the set point changes to match the new value of the condition.

C When a condition moves away from its set point, the body responds in a way that returns the condition to the set point.

D When a condition reaches its set point, the body responds in a way that moves the condition away from the set point.

9 · Part 2 Supplement Homeostasis [1]

Maria has not eaten for several hours. Her blood glucose concentration falls below the set point.

Which row shows the hormone that her pancreas secretes in larger amounts and the effect of this hormone on her liver?

	hormone secreted in larger amounts	**effect on the liver**
A	glucagon	glycogen is converted to glucose
B	insulin	glycogen is converted to glucose
C	glucagon	glucose is converted to glycogen
D	insulin	glucose is converted to glycogen

10 · Part 2 Supplement Homeostasis [1]

On a winter morning, a girl waits outside for a bus without a coat. Her body temperature starts to fall below the set point.

Which response helps to raise her body temperature?

A Her sweat glands produce more sweat.

B The arterioles supplying her skin surface capillaries dilate.

C Her hair erector muscles relax so that the hairs lie flat against the skin.

D Her skeletal muscles contract and relax rapidly and repeatedly.

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11 · Part 2 Supplement Homeostasis [1]

When the body is cold, vasoconstriction takes place in the skin.

Which statement explains how vasoconstriction reduces heat loss from the skin?

A The skin surface capillaries constrict and move deeper below the skin surface, so they are further from the air.

B The arterioles supplying the skin surface capillaries become narrower, so less blood flows through the skin surface capillaries and less heat is lost by radiation.

C The arterioles supplying the skin surface capillaries become wider, so blood flows through the skin surface capillaries more quickly and has less time to lose heat.

D The skin surface capillaries become narrower, so the sweat glands receive less blood and produce less sweat.

12 · Part 2 Core Tropic responses [1]

Some bean seedlings are laid on their sides in a dark cupboard. After two days, the roots have curved and are growing downwards.

Which row identifies the stimulus and the direction in which the roots grew in relation to it?

	stimulus	**direction of growth**
A	light	towards the stimulus
B	gravity	away from the stimulus
C	gravity	towards the stimulus
D	light	away from the stimulus

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13 · Part 2 Core Tropic responses [1]

Which observation is an example of phototropism?

A The root of a seedling grows downwards in a dark box.

B The shoots of a plant on a windowsill bend and grow towards the window.

C The stem of a climbing plant coils around a stick that it touches.

D The shoot of a seedling grows upwards in a box with no light.

14 · Part 2 Core Tropic responses [1]

In an investigation, four soaked pea seeds are pinned to damp cotton wool inside a closed, dark box. Each seed is pinned in a different position, so that the young roots first point upwards, downwards, left and right.

The box is left for three days.

Which statement describes how the roots and shoots grow?

A All the roots grow downwards and all the shoots grow upwards, whatever the starting position of the seed.

B The roots and shoots of each seed keep growing in the direction they were pointing at the start.

C All the roots grow upwards and all the shoots grow downwards, whatever the starting position of the seed.

D All the roots grow downwards and all the shoots grow towards the nearest side of the box.

15 · Part 2 Supplement Tropic responses [1]

The shoot of a young plant is lit from the left side only.

Which row describes the side of the shoot with the higher auxin concentration, what happens to the cells on that side, and the direction in which the shoot bends?

	side with higher auxin concentration	**effect on the cells on that side**	**direction of bending**
A	left side	elongate more	towards the left
B	right side	divide faster	towards the left
C	right side	elongate more	towards the right
D	right side	elongate more	towards the left

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16 · Part 2 Core Hormones [4]

Insulin is made in the pancreas. It passes into the blood and is carried to the liver, where it changes the activity of the liver cells.

(a) Describe what is meant by the term hormone. [3]

(b) The pancreas also makes the enzyme amylase. Amylase passes along a tube called a duct into the small intestine. It does not enter the blood.

Use your description in (a) to explain why amylase is **not** a hormone. [1]

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17 · Part 2 Core Hormones [6]

The table shows information about two responses in a person.

response	time from stimulus to start of response / s	duration of response / s
X: hand pulled away after touching a hot oven tray	0.2	0.5
Y: heart rate rises after a sudden fright	3	300

One response is controlled by the nervous system and the other is controlled by a hormone.

(a) Identify the response that is controlled by a hormone. Give **one** reason from the table for your answer. [2]

(b) Explain why the hormonal response takes longer to start than the nervous response. [2]

(c) Suggest **one** advantage of response X being fast **and one** advantage of response Y lasting a long time. [2]

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18 · Part 2 Supplement Hormones [6]

One student is sitting quietly in a classroom when a fire alarm sounds at 1 minute. She is frightened and secretes adrenaline. Her blood glucose concentration and heart rate are recorded.

time / min	blood glucose concentration / mg per 100 cm^3	heart rate / beats per minute
0	88	70
2	94	105
4	108	112
6	112	98
8	104	84
10	96	76

(a) Calculate the percentage increase in her heart rate between 0 minutes and 4 minutes.

Show your working. [2]

(b) Explain how the changes in blood glucose concentration and heart rate between 0 minutes and 4 minutes would help her if she needed to run from danger. [3]

(c) Name **one** other hormone, secreted by the pancreas, that also increases the blood glucose concentration. [1]

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19 · Part 2 Core Hormones [5]

One evening at dusk, a pet cat is sleeping in a garden when a large dog runs towards it. The cat's adrenal glands secrete adrenaline.

(a) State how adrenaline travels from the adrenal glands to its target organs. [1]

(b) The diameter of the cat's pupils increases.

Suggest **one** advantage of this to the cat at dusk. [1]

(c) The cat's breathing rate increases.

Suggest how this helps the cat to run away from the dog. [2]

(d) The cat's heart rate stays high for several minutes after the dog has gone.

Use your knowledge of hormonal control to explain why. [1]

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20 · Part 2 Core Homeostasis [4]

The blood glucose concentration of a healthy person was measured before and after a meal of rice. The person ate the meal at 0 minutes.

time / min	blood glucose concentration / mg per 100 cm^3
0	85
30	140
60	125
90	104
120	90
150	86

(a) Keeping the blood glucose concentration within narrow limits is an example of homeostasis.

Describe what is meant by homeostasis. [1]

(b) Calculate the increase in blood glucose concentration between 0 minutes and 30 minutes. Give the unit. [1]

(c) Suggest why the blood glucose concentration increased between 0 minutes and 30 minutes. [1]

(d) Name the hormone that caused the blood glucose concentration to decrease between 30 minutes and 120 minutes. [1]

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21 · Part 2 Supplement Homeostasis [5]

Body temperature is controlled by negative feedback.

An athlete rode an exercise bicycle hard from 0 minutes to 20 minutes and then rested. Her core body temperature was measured every 10 minutes.

time / min	core body temperature / °C
0	37.0
10	37.6
20	38.1
30	37.7
40	37.3
50	37.0
60	37.0

(a) Use the table to state the set point for her core body temperature. [1]

(b) Explain how negative feedback returned her core body temperature to the set point after she stopped exercising. [3]

(c) Suggest why her core body temperature continued to rise during the exercise, even though negative feedback was operating. [1]

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22 · Part 2 Supplement Homeostasis [6]

The table shows the concentrations of glucose, insulin and glucagon in the blood of a healthy person who did not eat for 10 hours after a meal.

time since meal / h	blood glucose concentration / mg per 100 cm^3	insulin concentration / arbitrary units	glucagon concentration / arbitrary units
2	92	14	20
4	88	9	32
6	84	6	48
8	83	5	61
10	83	5	66

(a) Describe how the concentrations of insulin and glucagon changed between 2 hours and 10 hours. [2]

(b) Explain how the increase in glucagon concentration stops the blood glucose concentration from falling any further. [3]

(c) Suggest why it is an advantage that the insulin concentration is low after 8 hours without food. [1]

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23 · Part 2 Supplement Homeostasis [5]

In people with Type 1 diabetes, the pancreas does not produce enough insulin.

(a) Outline the treatment of Type 1 diabetes. [3]

(b) Insulin is a protein.

Suggest why insulin is not taken as a tablet that is swallowed. [2]

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24 · Part 2 Supplement Homeostasis [6]

On a cold, windy day, a hiker's body temperature starts to fall below the set point.

(a) State the organ that detects the fall in blood temperature and coordinates the responses. [1]

(b) The hiker starts to shiver.

Explain how shivering helps to raise body temperature. [2]

(c) Many mammals, such as foxes, have thick fur. In cold weather, their hair erector muscles contract and the hairs stand up.

Explain how this reduces heat loss from the body. [2]

(d) Later, the hiker climbs a steep hill and sweats. At the top she stops to rest in the cold wind and quickly feels very cold.

Explain why her damp skin loses heat quickly. [1]

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25 · Part 2 Supplement Homeostasis [6]

Scientists measured the blood flow through the skin of a volunteer's fingertip at different air temperatures.

air temperature / °C	blood flow through skin of fingertip / arbitrary units
5	4
10	6
15	10
20	18
25	30
30	44
35	52

(a) Describe the relationship between air temperature and blood flow through the skin of the fingertip. [2]

(b) Explain how the blood flow through the skin is reduced at 5 °C, and how this helps to keep the body warm. [3]

(c) Explain how the high blood flow through the skin at 35 °C helps to keep the body temperature at the set point. [1]

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26 · Part 2 Core Tropic responses [4]

Plants respond to gravity and to light by changing their direction of growth.

(a) Describe what is meant by gravitropism. [1]

(b) Describe what is meant by phototropism. [1]

(c) When a seed germinates underground, its root grows downwards and its shoot grows upwards.

Suggest **one** advantage to the seedling of the root growing downwards **and one** advantage of the shoot growing upwards. [2]

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27 · Part 2 Core Tropic responses [5]

Some students grew cress seedlings in three boxes, P, Q and R, for three days.

- Box P had a small hole in one side wall, so light entered from one side only.
- Box Q was lit only by a lamp directly above the seedlings.
- Box R was completely dark.

They measured the mean angle by which the shoots bent away from the vertical.

box	mean angle of bending / °
P	42
Q	0
R	0

(a) Name the response shown by the shoots in box P. [1]

(b) Explain why the shoots in box Q did not bend. [1]

(c) The shoots in box R grew straight upwards. Name the response that caused this. [1]

(d) State **two** variables that the students should keep the same in all three boxes. [2]

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28 · Part 2 Core Tropic responses [5]

A clinostat is a piece of apparatus with a disc that turns slowly and continuously. Seedlings can be fixed to the disc.

Two young bean seedlings, X and Y, each with a short straight root and shoot, were fixed so that the root and shoot were horizontal.

- Seedling X was fixed to a clinostat that was switched on.
- Seedling Y was fixed to an identical clinostat that was switched off.

Both clinostats were kept in the dark for two days.

(a) Predict how the root and the shoot of seedling Y grew. [2]

(b) The root and shoot of seedling X kept growing horizontally.

Explain why. [2]

(c) Explain why the clinostats were kept in the dark. [1]

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29 · Part 2 Supplement Tropic responses [6]

Three groups of young grass shoots were lit from one side for two days.

group	treatment	growth after two days
1	no treatment	grew longer and bent towards the light
2	shoot tip cut off	did not grow longer and did not bend
3	shoot tip covered with a cap that does not let light through	grew longer but did not bend

(a) Explain the results for group 1. [4]

(b) Explain the results for group 2. [1]

(c) Suggest an explanation for the results for group 3. [1]

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30 · Part 2 Supplement Tropic responses [6]

One potted sunflower seedling was laid on its side in the dark. After one day, its shoot had started to curve upwards.

Scientists measured the auxin concentration and the mean cell length on the upper and lower sides of the curved part of the shoot.

side of shoot	auxin concentration / arbitrary units	mean cell length / µm
upper	35	40
lower	65	58

(a) Calculate the percentage by which the mean cell length on the lower side is greater than on the upper side.

Show your working. [2]

(b) Explain how the distribution of auxin causes the shoot to curve upwards. [3]

(c) Explain why this response is described as an example of the chemical control of plant growth. [1]

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